- Interpolacja Lagrange'a

$$W_n(x) = \sum_{j=0}^{n} y_j \prod_{\substack{k=0 \\ k\neq j}}^{n} \frac{(x - x_k)}{(x_j - x_k)}$$

- Interpolacja Newtona

I rząd:

$$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0}$$

$$[x_1, x_2] = \ldots$$

II rząd:

$$[x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0}$$

$$\vdots$$

$$W_n(x) = y_0 + [x_0, x_1]\cdot(x - x_0) + [x_0, x_1, x_2]\cdot(x-x_0)(x-x_1) + \\ + \ldots$$

- Aproksymacja

$$\sum_{i=0}^{n-1} \left(f(x_i) - \sum_{j=0}^{m-1} a_j x^j \right) x_i^k = 0$$

- Całkowanie (Newton-Cotes)

- wzór trapezów:

$$K(f) = \frac{b-a}{2}\left(f(a) + f(b)\right)$$

- wzór złożony trapezów

$$T_n = h\left(\frac{1}{2} f(x_0) + f(x_1) + f(x_2) + \ldots + \frac{1}{2} f(x_n)\right)$$

$$h = \frac{b-a}{n}$$

liczba węzłów: $n+1$

- wzór Simpsona

$$K(f) = \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right)$$

- wzór złożony Simpsona

$$S_n = \frac{h}{3}\Big[f(x_0) + 4\left(f(x_1) + f(x_3) + \ldots + f(x_{n-1})\right) + \\ + 2\left(f(x_2) + f(x_4) + \ldots + f(x_{n-2})\right) + f(x_n)\Big]$$

- Funkcje nieliniowe
  - metoda bisekcji

$$\langle a, b \rangle$$

$$x_1 = \frac{a+b}{2}$$

$$|f(x_1)| > \varepsilon \quad \text{robimy dalej}$$

$$f(a) \cdot f(x_1) > 0 \quad \vee \quad f(a) \cdot f(x_1) < 0$$

$$a = x_1 \,;\, b \qquad\qquad a \,;\, b = x_1$$

$$x_2 = \frac{a+b}{2}$$

  - metoda Newtona

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$|f(x_1)| > \varepsilon$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

- Metoda eliminacji Gaussa

$$\left[\begin{array}{ccc|c} a_{00} & a_{01} & \dots & b_1 \\ a_{10} & a_{11} & \dots & b_2 \\ \vdots & & & \end{array}\right]$$

I krok

$$W_1 = \frac{a_{10}}{a_{00}} \qquad a_{11} = a_{11}' - W_1 \cdot a_{01}$$

$$W_2 = \frac{a_{20}}{a_{00}} \qquad a_{12} = a_{12}' - W_1 \cdot a_{02}$$

$$W_3 = \frac{a_{30}}{a_{00}} \qquad a_{21} = a_{21}' - W_2 \cdot a_{01}$$

~~I~~ II krok

$$W_1 = \frac{a_{21}}{a_{11}} \qquad a_{21} = a_{21}' - W_1 \cdot a_{11}$$

$$W_2 = \frac{a_{31}}{a_{11}} \qquad a_{22} = a_{22}' - W_1 \cdot a_{12}$$

- Układ równań różniczkowych

$$\begin{cases} \dfrac{dy_1}{dt} = -(y_1 + y_2) \\ \dfrac{dy_2}{dt} = y_1 - y_2 \end{cases}$$

$y' = \frac{x}{y} \quad y(2) = 1$

$y' = f(x, y) \quad x_0 = 2 \quad y_0 = 1$

$y_1 = y_0 + h \cdot f(x_0, y_0) \quad x_1 = x_1 + h$

$y_2 = y_1 + h \cdot f(x_1, y_1)$

$$y_1(t = \text{nowe}) = y_1(t = \text{stare}) + \frac{dy_1}{dt} \cdot \Delta t$$

$$y_2(t = \text{nowe}) = y_2(t = \text{stare}) + \frac{dy_2}{dt} \cdot \Delta t$$

- Householder

$$A' = A - \frac{N \cdot \overbrace{N^T \cdot A}^{\alpha}}{\underbrace{\frac{1}{2} N^T \cdot N}_{\beta}}$$

1. $|A_{11}| = \sqrt{A^2 + B^2 + C^2}$
2. $N = A_{11} + \text{sign}(a_{11}) \cdot |A_{11}| \cdot e_1$
3. $\beta = \frac{1}{2} N \cdot N^T$
4. $\alpha_n = N^T \cdot A_n$
5. $A_n' = A_n - \frac{\alpha_n}{\beta} \cdot N$

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Ikrok

Gauss

$$A = \left[\begin{array}{cccc|c} -1 & -5 & 0,5 & 5,5 & 9,5 \\ -2 & 0 & -1 & 3 & -3 \\ -1,5 & -1,25 & 0,5 & -0,75 & -1,5 \\ 0 & -1,25 & 0,5 & 5,5 & 9,5 \end{array}\right]$$

$$\left[\begin{array}{ccc} a_{00} & a_{01} \dots & b_1 \\ a_{10} & a_{11} \dots & b_2 \\ \dots & & \vdots \\ \dots & & \end{array}\right]$$

$$A = \left[\begin{array}{cccc|c} -2 & 0 & -1 & 3 & -3 \\ -1 & -5 & 0,5 & 5,5 & 9,5 \\ -1,5 & -1,25 & 0,5 & -0,75 & -1,5 \\ 0 & -1,25 & 0,5 & 5,5 & 9,5 \end{array}\right]$$

$$W_1 = \frac{a_{10}}{a_{00}} = \cancel{\tfrac{1}{2}}\ 0,5$$

$$W_2 = \frac{a_{20}}{a_{00}} = 0,75$$

$$W_3 = \frac{a_{30}}{a_{00}} = 0$$

$$a_{11} = a_{11}' - W_1 \cdot a_{01} = -5 - \left(\tfrac{1}{2} \cdot 0\right) = -5$$

$$a_{12} = a_{12}' - W_1 \cdot a_{02} = 0,5 - \tfrac{1}{2} \cdot (-1) = 1$$

$$a_{13} = a_{13}' - W_1 \cdot a_{03} = 5,5 - \tfrac{1}{2} \cdot 3 = 4$$

$$b_1 = b_1' - W_1 \cdot b_0 = 9,5 - \tfrac{1}{2} \cdot (-3) = 11$$

$$a_{21} = a_{21}' - W_2 \cdot a_{01} = -1,25 - 0,75 \cdot 0 = -1,25$$

$$a_{22} = a_{22}' - W_2 \cdot a_{02} = 0,5 - 0,75 \cdot (-1) = 1,25$$

$$a_{23} = a_{23}' - W_2 \cdot a_{03} = -0,75 - 0,75 \cdot 3 = -3$$

$$b_2 = b_2' - W_2 \cdot b_0 = -1,5 - 0,75 \cdot (-3) = 0,75$$

$$A = \left[\begin{array}{cccc|c} -2 & 0 & -1 & 3 & -3 \\ 0 & -5 & 1 & 4 & 11 \\ 0 & -1,25 & 1,25 & -3 & 0,75 \\ 0 & -1,25 & 0,5 & 5,5 & 9,5 \end{array}\right]$$

III krok

$$W_1 = \frac{a_{21}}{a_{11}} = \frac{-1,25}{-5} = \frac{1}{4}$$

$$W_2 = \frac{a_{31}}{a_{11}} = \frac{-1,25}{-5} = \frac{1}{4}$$

$$A = \left[\begin{array}{cccc|c} -2 & 0 & -1 & 3 & -3 \\ 0 & -5 & 1 & 4 & 11 \\ 0 & -1,25 & 1,25 & -3 & 0,75 \\ 0 & -1,25 & 0,5 & 5,5 & 9,5 \end{array}\right]$$

$$a_{21} = a_{21}' - W_1 \cdot a_{11} = -1,25 - \frac{1}{4} \cdot (-5) = 0$$

$$a_{22} = a_{22}' - W_1 \cdot a_{12} = 1,25 - \frac{1}{4} \cdot 1 = 1$$

$$a_{23} = a_{23}' - W_1 \cdot a_{13} = -3 - \frac{1}{4} \cdot 4 = -4$$

$$b_2 = b_2' - W_1 \cdot b_1 = 0,75 - \frac{1}{4} \cdot 11 = -1,75$$

$$a_{31} = a_{31}' - W_2 \cdot a_{11} = -1,25 - \frac{1}{4} \cdot (-5) = 0$$

$$a_{32} = a_{32}' - W_2 \cdot a_{12} = 0,5 - \frac{1}{4} \cdot 1 = 0,25$$

$$a_{33} = a_{33}' - W_2 \cdot a_{13} = 5,5 - \frac{1}{4} \cdot 4 = 4,5$$

$$b_3 = b_3' - W_2 \cdot b_2 = 9,5 - \frac{1}{4} \cdot 11 = 7$$

$$A = \left[\begin{array}{cccc|c} -2 & 0 & -1 & 3 & -3 \\ 0 & -5 & 1 & 4 & 11 \\ 0 & 0 & 1 & -4 & -1,75 \\ 0 & 0 & 0,25 & 4,5 & 7 \end{array}\right]$$

HOUSEHOLDER

$$A' = A - \frac{N \cdot N^T}{\frac{1}{2} N^T \cdot N} \cdot A$$

α

β

1.

$$|A_{11}| = \sqrt{\sum_{i=1}^{n} a_{i1}^{2}} \qquad e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

2.

$$N = A_{11} + \operatorname{sign}(a_{11}) \cdot |A_{11}| \cdot e_1$$

$$\begin{bmatrix} a_{11} \cdot A_{11} + \operatorname{sign}(a_{11} \cdot |A_{11}|) \\ a_{12} \\ a_{13} \\ a_{14} \end{bmatrix}$$

3.

$$\beta = \frac{1}{2} N^T \cdot N$$

$$\begin{bmatrix} n_1 & n_2 & n_3 & n_4 \end{bmatrix} \cdot \begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \end{bmatrix}$$

$$\beta = \frac{1}{2} \left(n_1 \cdot n_1 + n_2 \cdot n_2 + n_3 \cdot n_3 + n_4 \cdot n_4 \right)^2$$

4. $\alpha_1 = N^T \cdot A_{11}$

$\alpha_2 = N^T \cdot A_{12}$

$\alpha_3 = N^T \cdot A_{13}$

$\alpha_4 = N^T \cdot B$

5.

$$A_{11}' = \begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \\ a_{14} \end{bmatrix} - \frac{\alpha}{\beta} \begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ n_4 \end{bmatrix}$$

Przykład.

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 2 & 3 \\ 2 & -1 & 3 & -2 & 0 \\ 0 & 1 & 2 & 1 & 1 \\ -2 & 3 & 1 & 0 & 2 \end{array}\right]$$

1. $|A_{11}| = \sqrt{1^2 + 2^2 + 0^2 + 2^2} = \sqrt{9} = 3$

2. $N = A_{11} + \operatorname{sign}(a_{11}) \cdot |A_{11}| e_1$

$$\begin{bmatrix} 1 \\ 2 \\ 0 \\ -2 \end{bmatrix} + \begin{bmatrix} 3 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 0 \\ -2 \end{bmatrix}$$

3. $\beta = \frac{1}{2} N \cdot N^T$

$$\beta = \frac{1}{2} \left[4\ 2\ 0\ {-2}\right] \cdot \begin{bmatrix} 4 \\ 2 \\ 0 \\ -2 \end{bmatrix} = \frac{1}{2}(16 + 4 + 0 + 4) = 12$$

4. $\alpha_1 = N^T \cdot A_{11} = \left[4\ 2\ 0\ {-2}\right] \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \\ -2 \end{bmatrix} = (4 + 4 + 0 + 4) = 12$

$$\alpha_2 = N^T \cdot A_{12} = \begin{bmatrix} 4 & 2 & 0 & -2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 8 - 2 + 0 - 6 \end{bmatrix} = 0$$

$$\alpha_3 = N^T \cdot A_{13} = \begin{bmatrix} 4 & 2 & 0 & -2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 3 \\ 2 \\ 1 \end{bmatrix} = -4 + 6 + 0 - 2 = 0$$

$$\alpha_4 = N^T \cdot A_{14} = \begin{bmatrix} 4 & 2 & 0 & -2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -2 \\ 1 \\ 0 \end{bmatrix} = 8 - 4 + 0 + 0 = 4$$

$$\alpha_B = N^T \cdot A_B = \begin{bmatrix} 4 & 2 & 0 & -2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 0 \\ 1 \\ 2 \end{bmatrix} = 12 + 0 + 0 - 4 = 8$$

5.

$$A_{11}' = A_{11} - \frac{\alpha_1}{\beta} \cdot N = \begin{bmatrix} 1 \\ 2 \\ 0 \\ -2 \end{bmatrix} - \frac{12}{12} \begin{bmatrix} 4 \\ 2 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A_{12}' = A_{12} - \frac{\alpha_3}{\beta} \cdot N = \begin{bmatrix} 2 \\ -1 \\ 1 \\ 3 \end{bmatrix} - \frac{0}{12} \begin{bmatrix} 4 \\ 2 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \\ 3 \end{bmatrix}$$

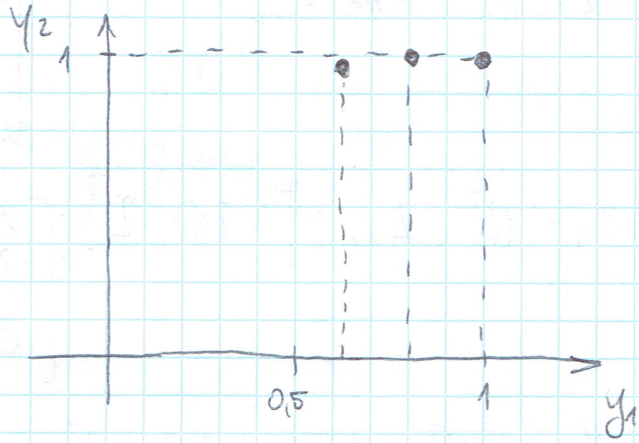
$A_{13}' = A_{13}$ bo $\alpha_3 = 0$

$$A_{14}' = A_{13} - \frac{\alpha_4}{\beta} \cdot N = \begin{bmatrix} 2 \\ -2 \\ 1 \\ 0 \end{bmatrix} \circ \frac{4}{12} \begin{bmatrix} 4 \\ 2 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ -\frac{8}{3} \\ 1 \\ \frac{2}{3} \end{bmatrix}$$

$$\left[\begin{array}{cccc|c} -3 & 2 & 1 & \frac{2}{3} & \frac{1}{3} \\ 0 & -1 & 3 & -\frac{8}{3} & -\frac{4}{3} \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 3 & 1 & \frac{2}{3} & \frac{10}{3} \end{array}\right]$$

Układ równań różniczkowych

$$\begin{cases} \frac{dy_1}{dt} = -(y_1 + y_2) \\ \frac{dy_2}{dt} = y_1 - y_2 \end{cases}$$



$y_1(t=0) = 1$
$y_2(t=0) = 1$
$\Delta t = 0{,}1s$

$\frac{dy_1}{dt} = -(1+1) = -2$

$\frac{dy_2}{dt} = 1 - 1 = 0$

$y_1(t=0{,}1) = y_1(t=0) + V_1 \cdot \Delta t = 1 + (-2) \cdot 0{,}1 = 0{,}8$
$y_2(t=0{,}1) = y_2(t=0) + V_2 \cdot \Delta t = 1 + 0 \cdot 0{,}1 = 1$

$y_1(t=0{,}1) = 0{,}8$
$y_2(t=0{,}1) = 1$
$\Delta t = 0{,}1s$

$\frac{dy_1}{dt} = -(0{,}8+1) = -1{,}8$

$\frac{dy_2}{dt} = 0{,}8 - 1 = -0{,}2$

$y_1(t=0{,}2) = y_1(t=0{,}1) + V_1 \cdot \Delta t = 0{,}8 + (-1{,}8) \cdot 0{,}1 = 0{,}62$
$y_2(t=0{,}2) = y_2(t=0{,}1) + V_2 \cdot \Delta t = \cancel{-0{,}2} + 1 + (-0{,}2) \cdot 0{,}1 = 0{,}98$

Rozkład Choleskiego

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ 2x_1 + 8x_2 + 10x_3 = 3 \\ 3x_1 + 10x_2 + 22x_3 = 7 \end{cases} \qquad A = L \cdot L^T$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 10 \\ 3 & 10 & 22 \end{bmatrix} * \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 7 \end{bmatrix} \qquad \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 10 \\ 3 & 10 & 22 \end{bmatrix}$$

$$A_1 = \det[1] = 1$$

$$A_2 = \det\begin{bmatrix} 1 & 2 \\ 2 & 8 \end{bmatrix} = 4$$

$$A_3 = \det\begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 10 \\ 3 & 10 & 22 \end{bmatrix} = \cancel{176 + 60 + 60 - 48 - 100 - 88} = 36$$

$$l_{11} = \sqrt{a_{11}} = 1$$

$$l_{21} = \frac{a_{21}}{l_{11}} = \frac{2}{1} = 2$$

$$l_{31} = \frac{a_{31}}{l_{11}} = \frac{3}{1} = 3$$

$$l_{22} = \sqrt{a_{22} - \sum_{k=1}^{1} (l_{2k})^2} = \sqrt{a_{22} - (l_{21})^2} = \sqrt{8 - 2^2} = 2$$

$$l_{32} = \frac{a_{32} - \sum_{k=1}^{1} l_{3k} \cdot l_{2k}}{l_{22}} = \frac{a_{32} - l_{31} \cdot l_{21}}{l_{22}} = \frac{10 - 3 \cdot 2}{2} = 2$$

$$l_{33} = \sqrt{a_{33} - \sum_{k=1}^{2} (l_{3k})^2} = \sqrt{a_{33} - (l_{31})^2 - (l_{32})^2} = 7$$

$$= \sqrt{22 - 3^2 - 2^2} = 3$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 2 & 3 \end{bmatrix}$$

$Ly = b$:

$$\begin{cases} y_1 = 1 \\ 2y_1 + 2y_2 = 3 \\ 3y_1 + 2 \cdot y_2 + 3y_3 = 7 \end{cases} \Rightarrow y = \begin{bmatrix} 1 \\ 0{,}5 \\ 1 \end{bmatrix}$$

$Lx = y$:

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ \quad 2x_2 + 2x_3 = 0{,}5 \\ \quad\quad 3x_3 = 1 \end{cases} \Rightarrow x = \begin{bmatrix} \frac{1}{6} \\ -\frac{1}{12} \\ \frac{1}{3} \end{bmatrix}$$

Odp.: $x = \left[\frac{1}{6}, -\frac{1}{12}, \frac{1}{3} \right]^T$

Interpolacja Lagrange'a

$$W_n(x) = \sum_{j=0}^{n} y_j \prod_{\substack{k=0 \\ k \neq j}}^{n} \frac{(x-x_k)}{(x_j-x_k)}$$

Przykład:

W punktach $-2, -1, 0, 2$ przyjmuje odpowiednio wartości $0, 1, 1, 2$

$n = 3$

$$W_3(x) = 0 \frac{(x+1)(x-0)(x-2)}{(-2+1)(-2-0)(-2-2)} +$$

$$+ 1 \frac{(x+2)(x-0)(x-2)}{(-1+2)(-1-0)(-1-2)} +$$

$$+ 1 \frac{(x+2)(x+1)(x-2)}{(0+2)(0+1)(0-2)} +$$

$$+ 2 \frac{(x+2)(x+1)(x-0)}{(2+2)(2+1)(2-0)} =$$

$$= \frac{1}{3}(x^3-4x) - \frac{1}{4}(x^3+x^2-4x-4) + \frac{1}{12}(x^3+3x^2+2x) =$$

$$= \boxed{\frac{1}{6}x^3 - \frac{1}{6}x + 1}$$

Przybliżona wartość w punkcie 1:

$$W_3(1) = \frac{1}{6} \cdot 1^3 - \frac{1}{6} \cdot 1 + 1 = 1$$

Interpolacja Newtona

Znajdź wielomian, który w punktach $-2, -1, 0, 2, 4$, odpowiednio przyjmuje wartości $-1, 0, 5, 99, -55$

$n = 4$

Ilorazy różnicowe I rzędu:

$$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0} = \frac{0-(-1)}{-1-(-2)} = 1$$

$$[x_1, x_2] = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5-0}{0-(-1)} = 5$$

$$[x_2, x_3] = \frac{y_3 - y_2}{x_3 - x_2} = \frac{99-5}{2-0} = 47$$

$$[x_3, x_4] = \frac{y_4 - y_3}{x_4 - x_3} = \frac{-55-99}{4-2} = -77$$

Ilorazy różnicowe II rzędu:

$$[x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0} = \frac{5-1}{0+2} = 2$$

$$[x_1, x_2, x_3] = \frac{[x_2, x_3] - [x_1, x_2]}{x_3 - x_1} = \frac{47-5}{2+1} = 14$$

$$[x_2, x_3, x_4] = \frac{[x_3, x_4] - [x_2, x_3]}{x_4 - x_2} = \frac{-77-47}{4-0} = -31$$

$$[x_0, x_1, x_2, x_3] = \frac{[x_1, x_2, x_3] - [x_0, x_1, x_2]}{x_3 - x_0} = \frac{14-2}{2+2} = 3$$

$$[x_1, x_2, x_3, x_4] = \frac{[x_2, x_3, x_4] - [x_1, x_2, x_3]}{x_4 - x_1} = \frac{-31-14}{4-(-1)} = -9$$

$$[x_0, x_1, x_2, x_3, x_4] = \frac{[x_1, x_2, x_3, x_4] - [x_0, x_1, x_2, x_3]}{x_4 - x_0} = \frac{-9-3}{4-(-2)} = -2$$

x_i	y_i	rząd I	rząd II	rząd III	rząd IV
-2	-1				
		1			
-1	0		2		
		5		3	
0	5		14		-2
		47		-9	
2	99		-31		
		-77			
4	-55				

dla $n=4$

$$W_4(x) = y_0 + [x_0, x_1]\cdot(x-x_0) + [x_0, x_1, x_2]\cdot(x-x_0)(x-x_1) +$$
$$+ [x_0, x_1, x_2, x_3]\cdot(x-x_0)(x-x_1)(x-x_2) +$$
$$+ [x_0, x_1, x_2, x_3, x_4]\cdot(x-x_0)(x-x_1)(x-x_2)(x-x_3)$$

$$W_4(x) = -2x^4 + x^3 + 19x^2 + 21x + 5$$

Aproksymacja metodą najmniejszych kwadratów

$$\sum_{i=0}^{n-1}\left(f(x_i) - \sum_{j=0}^{m-1} a_j x^j\right) x_i^k = 0$$

i	0	1	2	3	4
x_i	1	4	9	25	36
y_i	-6	-9	-12	-18	-21
0					
1					

1

$$(-6-(a_0\cdot 1^0+a_1\cdot 1^1))\cdot 1^0=-6-a_0-a_1$$
$$(-9-(a_0\cdot 4^0+a_1\cdot 4^1))\cdot 4^0=-9-a_0-4a_1$$
$$(-12-(a_0\cdot 9^0+a_1\cdot 9^1))\cdot 9^0=-12-a_0-9a_1$$
$$(-18-(a_0\cdot 25^0+a_1\cdot 25^1))\cdot 25^0=-18-a_0-25a_1$$
$$(-21-(a_0\cdot 36^0+a_1\cdot 36^1))\cdot 36^0=-21-a_0-36a_1$$

+

$$(-6-(a_0\cdot 1^0+a_1\cdot 1^1))\cdot 1^1=-6-1a_0-a_1$$
$$(-9-(a_0\cdot 4^0+a_1\cdot 4^1))\cdot 4^1=-36-4a_0-16a_1$$
$$(-12-(a_0\cdot 9^0+a_1\cdot 9^1))\cdot 9^1=-108-9a_0-81a_1$$
$$(-18-(a_0\cdot 25^0+a_1\cdot 25^1))\cdot 25^1=-450-25a_0-625a_1$$
$$(-21-(a_0\cdot 36^0+a_1\cdot 36^0))\cdot 36^1=-756-36a_0-1296a_1$$

+

Całkowanie

• wzór trapezów

$$K_{t}(f) = \frac{b-a}{2}\left(f(a) + f(b)\right)$$

$$\int_0^{2\pi} \cos^2 x + 1 \, dx \qquad h = \frac{\pi}{2}$$

$$n = \frac{b-a}{h} = \frac{2\pi - 0}{\frac{\pi}{2}} = 4$$

$$K(f) = \pi\left(\left[(\cos^2 0) + 1\right] + \left[(\cos^2 2\pi) + 1\right]\right)$$

$$K(f) = \pi(2+2) = 4\pi$$

• wzór złożony trapezów

$$T_n = h\left(\frac{1}{2} f(x_0) + f(x_1) + f(x_2) + \ldots + f(x_{n-1}) + \frac{1}{2} f(x_n)\right)$$

$$\int_0^1 \sqrt{1+x}\, dx \qquad h = \frac{1}{3}$$

$$n = \frac{b-a}{h} = \frac{1-0}{\frac{1}{3}} = 3$$

liczba węzłów: $n+1 = 4$

$x_0 = 0$ - lewy przedział całkowania

$x_1 = x_0 + h = \frac{1}{3}$

$x_2 = x_1 + h = \frac{2}{3}$

$x_3 = x_2 + h = 1$ - prawy przedział całkowania

$$T_3 = h\left(\frac{1}{2} f(x_0) + f(x_1) + f(x_2) + \frac{1}{2} f(x_3)\right) =$$

$$= \frac{1}{3}\left(\frac{1}{2}\sqrt{1} + \sqrt{1+\frac{1}{3}} + \sqrt{1+\frac{2}{3}} + \frac{1}{2}\sqrt{1+1}\right) \approx 1{,}2176$$

- wzór Simpsona

$$K(f) = \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right)$$

- wzór złożony Simpsona

$$S_n = \frac{h}{3}\Big[f(x_0) + 4\big(f(x_1) + f(x_3) + \ldots + f(x_{n-1})\big) +$$
$$+ 2\big(f(x_2) + f(x_4) + \ldots + f(x_{n-2})\big) + f(x_n)\Big]$$

Przykład.

$$\int_0^1 \sqrt{1+x}\, dx \qquad h = \frac{1}{4}$$

$$n = \frac{b-a}{h} = \frac{1-0}{\frac{1}{4}} = 4$$

węzły: $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$

$$S_4 = \frac{\frac{1}{4}}{3}\left[\sqrt{1} + 4\left(\sqrt{1+\frac{1}{4}} + \sqrt{1+\frac{3}{4}}\right) + 2\left(\sqrt{1+\frac{1}{2}}\right) + \sqrt{2}\right] =$$
$$\approx 1{,}21895$$

Funkcje nieliniowe

- metoda bisekcji

$$x^3 + x - 1 = 0 \qquad \langle a, b \rangle = \langle 0, 1 \rangle$$

$$\varepsilon = 0{,}01$$

Krok I:

$$a = 0 \; ; \; b = 1$$

$$x_1 = \frac{a+b}{2} = \frac{0+1}{2} = \frac{1}{2}$$

$$f(x_1) = f\left(\frac{1}{2}\right) = -0{,}375$$

$$|f(x_1)| > \varepsilon$$

~~krok II~~: $f(0) \cdot f\left(\frac{1}{2}\right) > 0$

krok II –

$$a = \frac{1}{2} \; ; \; b = 1$$

$$x_2 = \frac{\frac{1}{2}+1}{2} = \frac{3}{4}$$

$$f(x_2) = f\left(\frac{3}{4}\right) = 0{,}1782$$

$$f\left(\frac{1}{2}\right) \cdot f\left(\frac{3}{4}\right) < 0$$

krok III:

$$a = \frac{1}{2} \; ; \; b = \frac{3}{4}$$

$$|f(x_2)| > \varepsilon$$

$$x_3 = \frac{\frac{1}{2} + \frac{3}{4}}{2} = \frac{5}{8}$$

$$f(x_3) = f(0{,}625) = -0{,}131$$

$$f(x_3) \cdot f(a) > 0$$

$$f(0{,}625) \cdot f\left(\frac{1}{2}\right) > 0$$

$$a = \frac{5}{8} \qquad b = \frac{3}{4}$$

- **Metoda Newtona**

$y = e^x + 2x + 1$ $\qquad \varepsilon = 1e-5$

$y' = e^x + 2$

$x_0 = 0$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0 - \frac{2}{3} = -0{,}666667$$

~~x_2~~ $f(x_1) = 0{,}18008379$

$|f(x_1)| > 1e-5$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = -0{,}738316$$

Kwadratura Gaussa - Czebyszewa

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_n(x) = 2x\,T_{n-1}(x) - T_{n-2}(x)$$

Funkcja wagowa:
$$\omega(x) = \frac{1}{\sqrt{1-x^2}}$$

przedział całkowania: $[-1, 1]$

$$\int_{-1}^{1} f(x) \frac{1}{\sqrt{1-x^2}}\,dx \approx \sum_{i=0}^{n} A_i f(x_i) + R$$

x_i - pierwiastki wielomianu ortogonalnego stopnia $(n+1)$

A_i - odpowiadające współczynniki x_i

Wartości węzłów:
$$x_i = \cos\left(\frac{2i-1}{2n}\pi\right), \quad A_i = \frac{\pi}{n}$$

Przykład:
$$\int_{-1}^{1} \frac{\ln(x+3)}{\sqrt{1-x^2}}\,dx \qquad n = 4$$

$$\int_{-1}^{1} f(x) \frac{1}{\sqrt{1-x^2}}\,dx \approx \frac{\pi}{n}\sum_{i=1}^{n} f(x_i) + R$$

$$x_i = \cos\left(\frac{2i-1}{2n}\pi\right); \quad A_i = \frac{\pi}{n}$$

$$x_1 = \cos\left(\frac{2-1}{8}\pi\right) = \cos\frac{\pi}{8}$$
$$x_2 = \cos\left(\frac{4-1}{8}\pi\right) = \cos\frac{3}{8}\pi$$
$$x_3 = \cos\left(\frac{6-1}{8}\pi\right) = \cos\frac{5}{8}\pi$$
$$x_4 = \cos\left(\frac{8-1}{8}\pi\right) = \cos\frac{7}{8}\pi$$

$$f(x) = \ln(x+3)$$
$$n = 4$$

$$\frac{\pi}{4}\sum_{i=1}^{n} \ln(x_i+3)$$

$$\frac{\pi}{4}\cdot\left(\ln\left(\cos\frac{\pi}{8}+3\right) + \ln\left(\cos\frac{3}{8}\pi+3\right) + \ln\left(\cos\frac{5}{8}\pi+3\right) + \right.$$
$$\left. + \ln\left(\cos\frac{7}{8}\pi+3\right)\right) + R \qquad R = \frac{2\pi}{2^8(8!)}f^{(8)}$$

Metoda Rungego - Kutty

$$k_1 = h \cdot f(x_n, y_n)$$
$$k_2 = h \cdot f(x_n + 0{,}5h, y_n + 0{,}5k_1)$$
$$k_3 = h \cdot f(x_n + 0{,}5h, y_n + 0{,}5k_2)$$
$$k_4 = h \cdot f(x_n + h, y_n + k_3)$$
$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4), \quad n = 0, 1, \ldots, N;$$

$$y' = 2x + y \; ; \; y(0) = 1 \; ; \; h = \frac{1}{10}$$

$$y' = f(x, y) \; ; \; x_0 = 0 \; ; \; y_0 = 1$$

$$k_1 = \frac{1}{10} \cdot 1 = \frac{1}{10}$$
$$k_2 = \frac{1}{10} \cdot f\left(\frac{1}{20}, 1 + \frac{1}{20}\right) = \frac{23}{200}$$
$$k_3 = \frac{1}{10} \cdot f\left(\frac{1}{20}, 1 + \frac{23}{400}\right) = \frac{1}{10} \cdot \left(\frac{2}{20} + \frac{423}{400}\right) = \frac{463}{4000}$$
$$k_4 = \frac{1}{10} \cdot f\left(\frac{1}{10}, 1 + \frac{463}{4000}\right) = \frac{1}{10} \cdot f\left(\frac{1}{10}, \frac{4463}{4000}\right) = \frac{1}{10}\left(\frac{2}{10} + \frac{4463}{4000}\right) =$$
$$\frac{1}{10} \cdot \frac{5263}{4000} = \frac{5263}{40000}$$

$$y_{n+1} = 1 + \frac{1}{6}\left(\frac{1}{10} + 2 \cdot \frac{23}{200} + 2\frac{463}{4000} + \frac{5263}{40000}\right)$$

Szacowanie błędów

$$\frac{1}{1+\varepsilon} \approx (1-\varepsilon) \ ; \ \sqrt{1+\varepsilon} = 1 + \frac{\varepsilon}{2} \ ; \ (1+\varepsilon_1)(1+\varepsilon_2) = (1+\varepsilon_1+\varepsilon_2)$$

$$x^2 + y^2 = 2 \cdot 2^{-t} \qquad \sqrt{} = \frac{3}{2} \cdot 2^{-t}$$

$$\left.\begin{matrix} x+y \\ x-y \\ x \cdot y \\ x/y \end{matrix}\right\} 2^{-t}$$

$$\omega = \frac{x^4 - y^4}{\sqrt{x^2+y^2}} = \frac{(x^2-y^2)(x^2+y^2)}{\sqrt{x^2+y^2}} = \frac{(x^2-y^2)\cancel{(x^2+y^2)}\sqrt{x^2+y^2}}{\cancel{(x^2+y^2)}} =$$

$$= (x^2-y^2)\sqrt{x^2+y^2} = (x-y)(x+y)\sqrt{x^2+y^2} =$$

$$= [(x-y)(x+y)](1+\varepsilon_1) \cdot \sqrt{(x^2+y^2)(1+\frac{\varepsilon_2}{2})}(1+\varepsilon_3)(1+\varepsilon_4) =$$

$$= \left[(x-y)(x+y)\sqrt{x^2+y^2}\right]\left(1+\varepsilon_1+\frac{\varepsilon_2}{2}+\varepsilon_3+\varepsilon_4\right) \leq$$

$$\omega\left(1+\varepsilon_1+\frac{\varepsilon_2}{2}+\varepsilon_3+\varepsilon_4\right) = \omega(1+\varepsilon)$$

ε_1 - $(x-y)(x+y) = 3 \cdot 2^{-t}$

ε_2 - $x^2+y^2 = 2 \cdot 2^{-t}$

ε_3 - $\sqrt{} = \frac{3 \cdot 2^{-t}}{2}$

ε_4 - mnożenie $= 2^{-t}$

$$|\varepsilon| \leq \left|\varepsilon_1 + \frac{\varepsilon_2}{2} + \varepsilon_3 + \varepsilon_4\right|$$

$$|\varepsilon| \leq |\varepsilon_1| + \left|\frac{\varepsilon_2}{2}\right| + |\varepsilon_3| + |\varepsilon_4|$$

$$|\varepsilon| \leq 3 \cdot 2^{-t} + 2^{-t} + \frac{3 \cdot 2^{-t}}{2} + 2^{-t}$$

$$|\varepsilon| \leq 6{,}5 \cdot 2^{-t}$$